

1. Nicotine (C) in water (A) solution containing 1.0 % nicotine is to be extracted with kerosene (B). Water and Kerosene are completely insoluble. Determine the percentage extraction of nicotine if 100 kg of feed solution is extracted in (a) One single stage with 150 Kg solvent; (b) three stages in cross flow configuration. All the stages use 50 Kg of kerosene.

(C) Same feed is now extracted in a counter current system. The target is to reduce the Nikotene content to 0.1%. Determine the number of stages if 115 KG of kerosene is used.

Equilibrium data:

x' (kg nicotine / kg water)	0	0.001011	0.00246	0.00502	0.00751	0.00998	0.0204
y'^* (kg nicotine / kg kerosene)	0	0.000807	0.001961	0.00456	0.00686	0.00913	0.0187